

Problem Statement 4

Title of Problem Statement

Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals

Description (Background, Detailed Description and Expected Solution):

Background:

Free Space Optical Communication (FSOC) offers unprecedented advantages for next-generation mobile networks, including gigabit-to-terabit data rates, license-free spectrum operation, high immunity to electromagnetic interference, etc. However, deploying FSOC links between mobile platforms (satellites, UAVs) presents a severe challenge of pointing, acquisition and tracking (PAT) of highly narrow laser beams. PAT typically happens in two stages: coarse alignment and fine alignment. Coarse alignment is one of the key challenges of PAT, where the transmitting terminal must first locate and maintain the remote terminal within its camera Field-of-View (FOV).

Developing and testing such algorithms on real hardware requires expensive cameras, pan-tilt mechanisms, and optical components & equipment. A software based virtual camera tracking provides an inexpensive and accessible platform for algorithm development and learning.

Description:

Unlike conventional radio-frequency systems, FSOC relies on a highly directional optical beam. Even a small angular error can prevent successful communication. Before fine pointing mechanism can take over, a coarse alignment stage must:

- Observe the surrounding environment,
- Acquire and detect the remote terminal or beacon,
- Estimate the position, and
- Continuously adjust the pointing direction to maintain visibility.

The participants shall develop this coarse alignment process in software, allowing to develop and validate tracking algorithms without specialized hardware and setup. The following section provides reference parameters and performance criteria to be considered for the software development.

Parameters and Specifications:

Functional Objective: Develop a software system that autonomously detects, identifies, and continuously tracks a designated moving target within a virtual scene by controlling a virtual camera viewport.

Sr. No.	Parameter	Suggested Value	Remarks
Camera Parameters			
1.	Screen Size (min.)	2000 x 2000 pixels	Optional: User-defined
2.	Camera Type	Monochrome, Focal Plane Array	Optional: Colour
3.	Camera Resolution	640 x 480 pixels	Optional: User-defined
4.	Camera FOV	User-defined	Default: 4° x 3°
5.	Camera update Rate	30 Hz (min.)	
6.	Initial Camera Position	Centre of the Screen	
Target Parameters			
7.	Target Type	Beacon Spot	
8.	Number of Targets	1, mandatory	multiple optional
9.	Target Shape	User-defined	Default: Square
10.	Target Size	5-20 x 5-20 pixels (user-defined)	Default: 10 x 10
11.	Initial Target Location	User-defined	Default: Random
12.	Motion	Selectable, at least four: Straight Line, Circular, Figure of 8, Random	Optional: Spiral, Sinusoidal, User-defined
Camera Motion Constraints			
13.	Max. Pan Speed	5-10 °/s (User-defined)	Default: 5 °/s
14.	Max. Tilt Speed	5-10 °/s (User-defined)	Default: 5 °/s
15.	Update Interval	≥ 20 Hz	
Performance Specifications			
16.	Acquisition Time	≤ 2 sec	
17.	Tracking Error	≤ 10 pixels	
18.	Target Loss	< 5%	

Sr. No.	Parameter	Suggested Value	Remarks
19.	Re-acquisition Time	≤ 1 sec	
20.	Processing Speed	≥ 20 FPS	
Disturbances and Noise			
21.	Image Noise	1. Salt & Pepper (around 10% of image), 2. Gaussian & 3. Poisson	User Selectable (one or more)
2.	Max. Standard Deviation of Noise	20 pixels	User-defined
3.	Max. Camera Jitter	± 20 pixels / frame	User-defined
4.	Atmospheric Disturbance	Clear, Haze, Fog, Rain, Low light	User-defined reduction in contrast and brightness
5.	Platform Motion	± 20 pixels/frame (max.)	User selectable Default/Mandatory: Linear Optional: Circular, random, spiral, figure of 8, etc.

Key Milestones:

Expected Solution:

Participants shall develop an AI-assisted camera tracking system capable of automatically detecting and continuously tracking a moving optical beacon in a simulated video stream while controlling a virtual pan-tilt camera.

The developed software shall be able to:

- Generate a configurable virtual environment,
- Generate one or more moving targets,
- Implement a movable virtual camera,
- Detect the target beacon automatically,
- Track the beacon continuously using computer vision,
- Control and reposition the virtual camera,
- Generate and introduce disturbances due to atmospheric turbulence, platform vibrations, camera motion, noise, etc., in the virtual camera feed,
- Display tracking performance and statistics in real-time

Deliverables

Each participating team shall submit the following mandatory deliverables:

Software Application

A standalone executable application implementing the complete virtual camera tracking system. The application shall provide all the mandatory functions and features as described above.

Source Code

Complete source code with proper documentation. The code shall be modular and adequately commented.

Technical Report

The technical report (about 10-15 pages) containing problem understanding, system architecture, description of software modules, tracking methods, AI methods (if used), test methodology, performance analysis and future improvements shall be submitted.

User Manual

The user manual with the description of installation of software, application operation, parameter configuration, GUI description, etc. shall be submitted. A 3–5 minutes video may also be provided as an optional deliverable for demonstration of the application.

Performance Log

The software should be capable of automatically generating a performance report containing simulation duration, FPS, acquisition time, average and maximum tracking error, lock retention rate, processing time, etc.

Evaluation Method and Criteria

The solutions developed by participating teams will be evaluated using multi-layered evaluation method. The following table describes stages of evaluation, their weightage and methods.

Evaluation Stage	Description	Evaluation Method and Criteria	Marks (%)
Functional Verification	Teams are given 10-15 minutes to demonstrate the software and its functionality.	1. Implementation of all mandatory functions 2. Operational success	20

Evaluation Stage	Description	Evaluation Method and Criteria	Marks (%)
Benchmark Performance-1	Each team will be given few scenarios.	3. GUI 1. Execution of the scenario 2. Log of Centroiding error 3. Automatically generated performance logs	30
Benchmark Performance-2	Each team will be given a few video files (.mp4) @30 fps, covering a complete screen with noise and moving beacon spot. The software needs to bypass its PTZ camera and take this video as an input to the coarse pointing system.	1. Comparison of Centroiding error with predefined error values 2. Performance w.r.t. various parameters like RMSE, acquisition and re-acquisition time, Lock retention rate, FPS, etc.	30
Technical Evaluation	Teams present their approach, methods, architecture, design, etc to the evaluators.	1. Understanding of the problem 2. System architecture and software design 3. Selection of Algorithms 4. AI and computer vision 5. Innovation and Novelty 6. Technical Documentation and Presentation 7. Technical Discussion and Q&A	20

Organization (Ministry/Department Name): Department of Space / Indian Space Research Organisation

Category (Hardware/Software): Software

Theme (Available at www.sih.gov.in): Smart Automation, Space Technology

Youtube Link/Video Link: NA

Dataset Link (real-data or dummy data): NA